

IEEE VR 2025 Workshops: 3rd Workshop on XR Technologies for Healthcare, Wellbeing, and Medicine

Joaquim Jorge University of Lisbon, Portugal	Matias Volonte Clemson University, USA	Craig Anslow Victoria University of Wellington, New Zealand
Andrew Duchowski Clemson University, USA	Timothy Bickmore Northeastern University, USA	Benjamin C. Lok University of Florida, USA
Catarina Pinto Moreira University of Technology Sydney, Australia	Anderson Maciel University of Lisbon, Portugal	Wolfgang Hürst Utrecht University, Netherlands
Daniel Roth Technical University of Munich, Germany	Danny Schott University of Magdeburg, Germany	Florian Heinrich* University of Magdeburg, Germany



ABSTRACT

The 3rd edition of the IEEE VR XR Health workshop aims to connect international researchers in medical, health, and wellbeing XR applications. This recurring event is intended to give young researchers a stage to present their earlier works and connect them with more experienced researchers in the field. With 23 workshop paper contributions, one keynote, and a panel discussion, we are striving to advance the development of an international healthcare, wellbeing, and medical XR community.

Index Terms: Healthcare, Wellbeing, Medical XR

1 MESSAGE FROM THE ORGANIZERS

Building on the success of previous editions, XR Health 2025 emphasizes participation, discussion, and the establishment of long-term research connections. The workshop welcomes early-stage and mature research contributions, supporting an inclusive environment bridging academic and applied research perspectives. While the event is primarily in-person, the keynote and paper presentations are available remotely to ensure broader engagement. We thank all contributors, reviewers, and attendees who made this event possible. Special thanks go to our invited speaker, whose clinical insights undoubtedly enrich the discourse on XR's role in modern healthcare. As the field continues to evolve, we hope that the XR

*e-mail: florian.heinrich@ovgu.de

Health 2025 workshop inspires new directions in research and fosters collaborations that drive the future of XR in medicine and wellbeing. We look forward to the discussions and discoveries ahead.

2 ABOUT THE WORKSHOP

The XR Health 2025 workshop, in conjunction with IEEE VR 2025, marks the third iteration of a dedicated forum exploring the transformative impact of Extended Reality (XR) technologies on healthcare, well-being, and medicine. Following the success of previous editions, this workshop continues to serve as a key venue for researchers, clinicians, and industry professionals to discuss innovative applications of XR in healthcare and address emerging challenges in the field. With the increasing adoption of XR technologies—encompassing Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR)—the healthcare sector witnessed significant advancements in clinical therapy, rehabilitation, surgery, medical training, and patient well-being. However, despite the promising potential of XR for medical applications, critical research questions remain regarding its efficacy, accessibility, and integration into existing healthcare workflows. XR Health 2025 provides a platform for interdisciplinary dialogue, fostering collaboration between the Human-Computer Interaction (HCI), medical, and XR communities. This year, the workshop features a series of paper presentations, a keynote talk from the medical domain, and interactive breakout sessions designed to facilitate knowledge exchange and networking. This is further supplemented by poster presentations of accepted submissions.

3 SCOPE

We invited contributions exploring the application of XR technology within the broader contexts of health, wellbeing, and medicine. Suggested topics included, but were not limited to:

- XR for Surgery and Diagnostics: Applications in minimally invasive interventions, surgical procedures, and diagnostic imaging, including cardiac, general, dental, or orthopedic surgery; MRI, CT, PET, or X-ray imaging; and radiation therapy.
- XR for Clinical Therapy: Applications enhancing therapeutic interventions in clinical settings, such as patient safety, clinical care, mental health treatment (e.g., depression, PTSD, stress, anxiety), pain management (e.g., chronic pain, acute pain), and addiction treatment (e.g., support for recovery processes).
- XR for Rehabilitation and Supportive Care: Applications facilitating patient recovery and ongoing support, including rehab and physical activity, cancer pain and palliative care, neurodevelopmental disorders (e.g., autism, ADHD), nursing and midwifery, and long-term recovery support.
- XR for Remote Care: Applications enabling remote healthcare delivery and emergency medical services, such as telehealth solutions and emergency medical care.
- XR for Training and Education: Applications enhancing the education and training of healthcare professionals, including clinical skills development, surgical simulations, and immersive learning experiences.
- XR for Wellbeing: Applications promoting wellness and preventive health, such as stress management, mental health support, and tools for enhancing overall wellbeing.
- XR for Chronic Disease Management: Applications to promote home-based health education and health behavior change interventions, particularly for chronic disease self-care.

4 CONTRIBUTIONS

The XR Health workshop at IEEE VR 2025 showcases 23 papers exploring the integration of XR technologies in healthcare, rehabilitation, and medical training. Key contributions include VR-based physical therapy guidance, augmented and diminished reality for cognitive training, and XR-enhanced surgical training and diagnostics. Several papers examine remote medical teleconsultation, VR exergames for elderly balance training, and markerless AR visualization of vascular structures. The research highlights the growing impact of XR on patient care, mental health interventions, and clinical education, reinforcing its potential to transform healthcare practices through innovative, immersive solutions. The full list of articles includes the following contributions:

- *A Survey of Emotional Intelligence Applications In Immersive And Non-Immersive Technologies* – Heidarikohol et al.
- *An Augmented and Diminished Reality Exergame for Investigating the Embodiment of a Prosthetic Limb* – Macaluso et al.
- *Applying Multimodal Mixed Reality System for Classifying Parkinson's Disease: Design and Evaluation of the Voice Module* – Stepień et al.
- *CLEAR: Colon Segmentation with Liquid Error Avoidance and Refinement for VR Colonography* – Freire et al.
- *Co-design of virtual reality exergames for balance rehabilitation* – Toftgaard Henriksen et al.
- *Evaluating VR and AR Mirror Exposure for Anorexia Nervosa Therapy in Adolescents: A Method Proposal for Understanding Stakeholder Perspectives* – Fiedler et al.

- *Extended Reality Teleconsultation for Medical Interventions: State of the Art and Perspectives* – Schieber et al.
- *Immersive Mixed Reality Training Concept for Mastering Surgical Knot-tying* – Queisner et al.
- *Impact of Input and Output Devices on a Virtual Ultrasound Training* – Allgaier et al.
- *Impact of Real-Time Feedback in Virtual Aseptic Technique Simulator for Basic Nursing Education* – Morishita et al.
- *Integrated Telehealth and Extended Reality to Enhance Home Exercise Adherence Following Total Hip and Knee Arthroplasty* – Conroy et al.
- *Integrating Virtual Reality and Physiological Measures to study Restorative Environments for Mental Well-Being* – Farmahini Farahani et al.
- *LARACROFT: Pilot test of Laparoscopy with Augmented Reality Adaptations for a Collaborative Framework* – Salgueiro et al.
- *Legato: Virtual Reality for Physical Rehabilitation of Patients in the Intensive Care Unit* – Eom et al.
- *Markerless Registration and Visualization of Volumetric Medical Data for Enhanced Surgical Precision Using Head-Mounted Displays* – Jurgas et al.
- *Mixed Reality 6D Object Pose Estimation using Deep Learning for Visual Markerless Surgical Navigation* – Trojak et al.
- *Prophylactically Limiting Phantom Limb Pain using Collaborative Virtual Reality* – Johnsen et al.
- *Refining Functional Vision Assessment: Challenges in Adapting Orientation and Mobility Tests to Virtual Reality* – Huang et al.
- *Supporting Shared Decision-Making Between Patients and Physicians: Using Hybrid VR Interface for Collaborative MRI Visualization* – Cybulski et al.
- *Virtual Reality Meets Machine Learning: Predicting HRV and Stress Outcomes for Mental Well-Being* – Kumar et al.
- *VR rehabilitation: A pilot study of HMD-based guided exercise & rehabilitation in VR for post-shoulder-surgery patients at home* – Guldager et al.
- *XR for educational anatomy: an Immersive human atlas with chatbot* – Uhl et al.
- *XR for Mental Health: How Users Perceive New Technologies?* – Figueroa et al.

ACKNOWLEDGMENTS

The XR Health chairs thank all the paper authors who submitted and presented their work at the workshop. We also thank our invited speaker and the IEEE VR 2025 Organizing Committees, particularly the Workshop Chairs: Rebecca Fribourg, Hugo Brument, Dirk Reiners, and Carolin Wienrich.